CONTOSO DATA DASHBOARD PROJECT

IEC - DATA ANALYTICS - COHORT 25

PRESENTED BY SHEEMA AHMED, SYED HAMMAD ATIF, SYEDA WASSAMA ALI & TAROOB ANWER

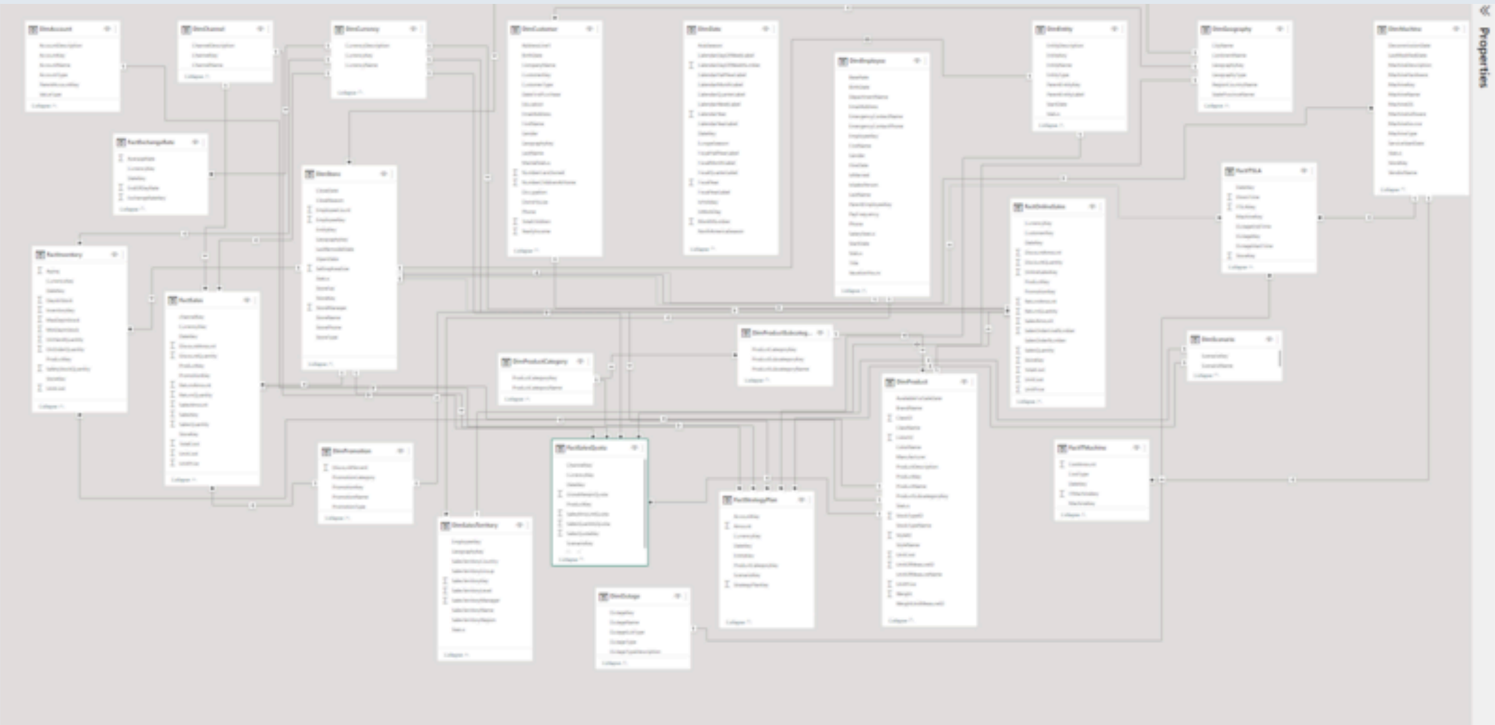
What is Contoso?

- A fictional multinational retail company
- Dataset represents realistic retail operations
- Contains transaction-level sales data across regions and time
- Designed to support business decision-making analysis

Learning Outcomes

- Develop practical skills in SQL, Excel, and Power BI
- Clean and analyze large datasets effectively
- Design interactive dashboards
- Interpret data to identify business patterns/trends
- Present insights for business decision-making

Dataset Overview



- **Fact Tables**
- Central tables that store measurable values
- Represent events or transactions
- Used for counts, totals, and trends

- **Dimension Tables**
- Provide descriptive context
- Contain attributes like who, what, when, where
- Used for filtering and grouping

25 tables

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Business Questions

Customer Insight

Which customers are at risk of churning?

What is the average order value (AOV) per customer?

Who are the top customers by revenue?

Sales Trends

Which regions or stores have the highest sales?

What are the seasonal trends in sales?

Which products are frequently bought together?

What is the return rate per product category?

Opperational Efficiency

Which products generate the highest revenue?

Which products are frequently bought together?

What is the return rate per product category?

Which products have the highest profit margins?

How is the trend in profit margins over time?

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Analytics Workflow

- Verified SQL outputs
- Checked for:
- Outliers, Missing values, Inconsistent records
- Used as a quality-control layer

- Imported large datasets efficiently
- Joined fact and dimension tables
- Calculated KPIs and different types of analyses
- Ensured accurate and scalable analysis

- Used for Visualization & Insights
- Built interactive dashboards
- Translated numbers into business insights with visuals

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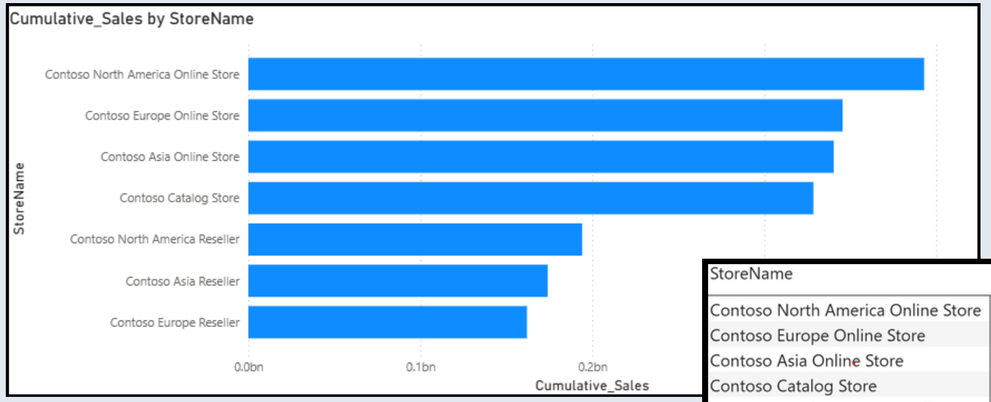
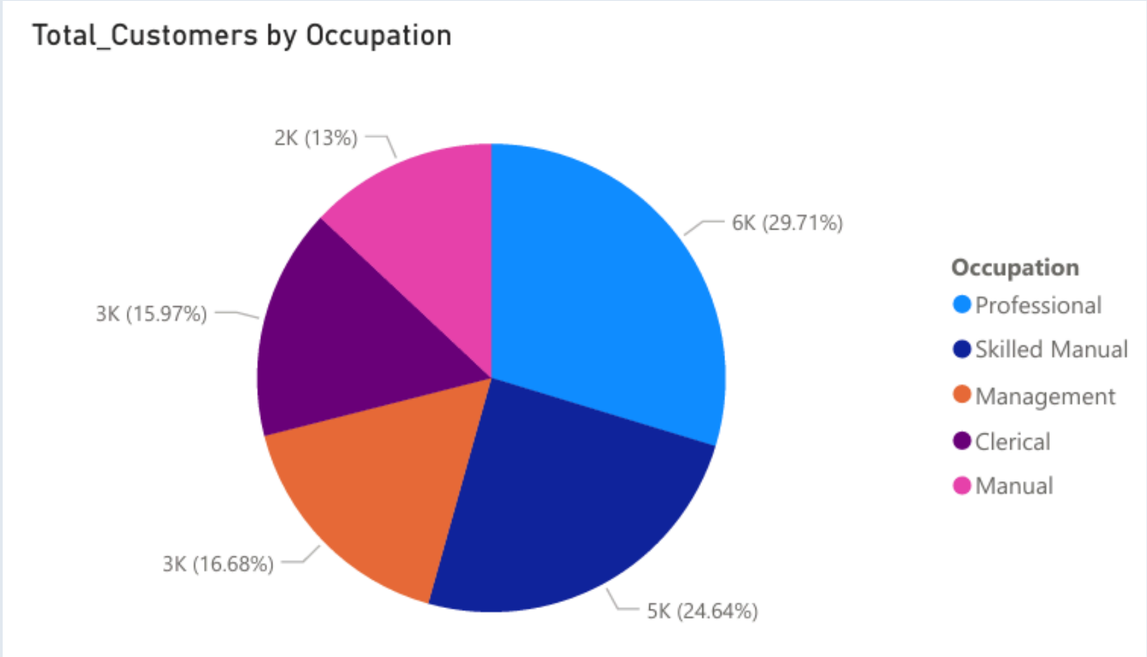
CUSTOMER INSIGHTS

Sales Across Stores



StoreName	Cumulative_Sales
Contoso North America Online Store	392,896,688.04
Contoso Europe Online Store	345,463,576.00
Contoso Asia Online Store	340,332,155.62
Contoso Catalog Store	328,507,425.71
Contoso North America Reseller	194,057,925.33
Contoso Asia Reseller	174,066,223.35
Contoso Europe Reseller	161,973,205.56

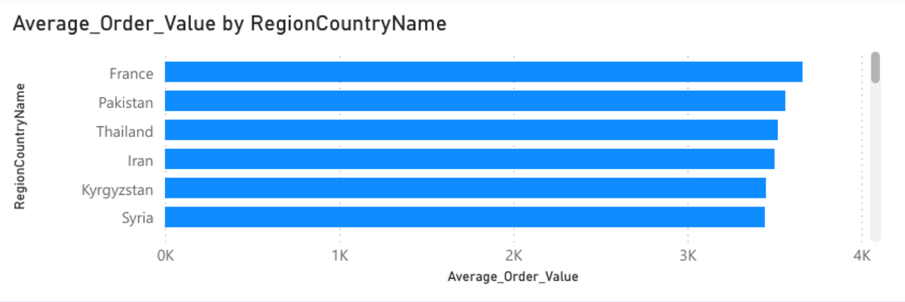


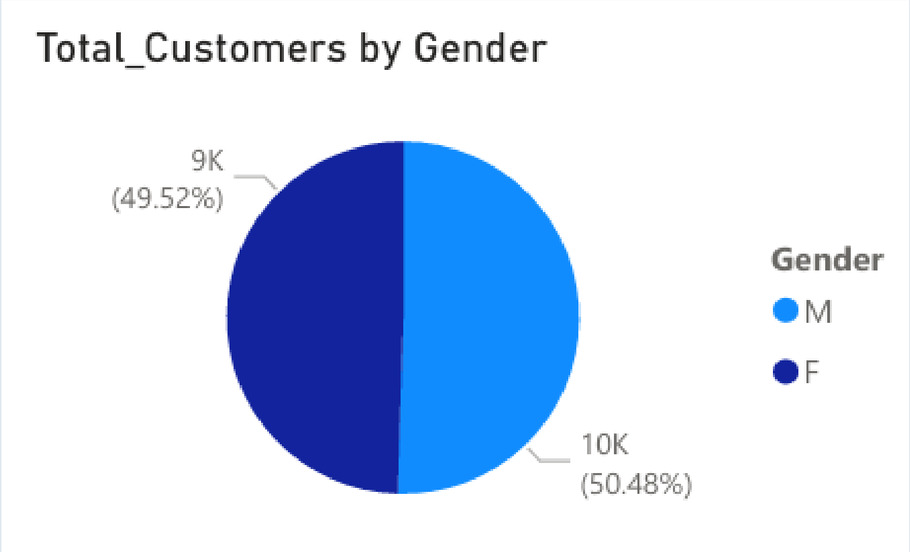
Total Customers

19K

1.94K

AOV per customer





Customer Profile vs AOV: Professional customers drive higher average order value.
Gender Distribution: Even split → gender-neutral targeting is effective.
Regional AOV vs Sales: Some regions show high AOV with low volume
Store Sales: Online stores have highest sales

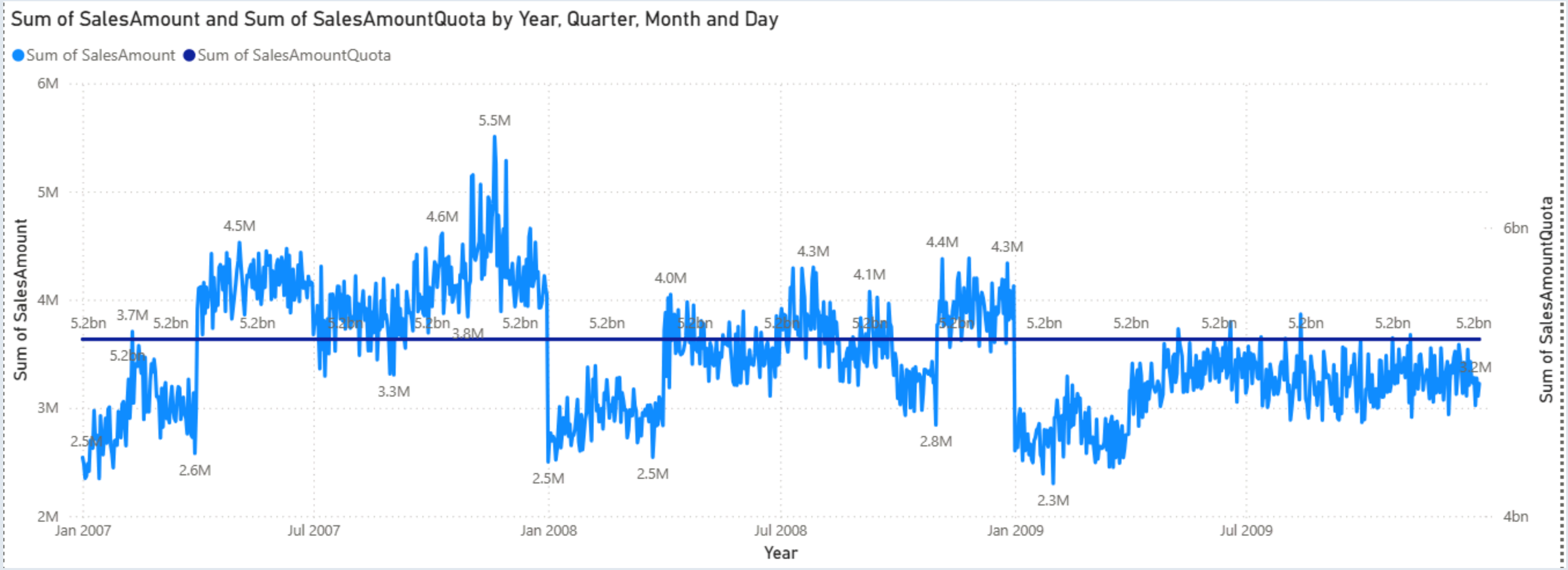
SALES TRENDS

4.07bn	17M	2M
Total_Sales	Total_Quantity	Total_Orders

259.09M	3.81bn
Total_Online_Sales	Total_Store_Sales

- Sales grew well in 2007, often hitting or passing the target.
- Early 2008 saw a big drop, then sales stayed near the target but with ups and downs.
- In 2009, sales mostly stayed below the target and dropped again.
- Overall, sales got more unstable and weaker over time.

Sales Trends Across Years with Sales Target as refernce



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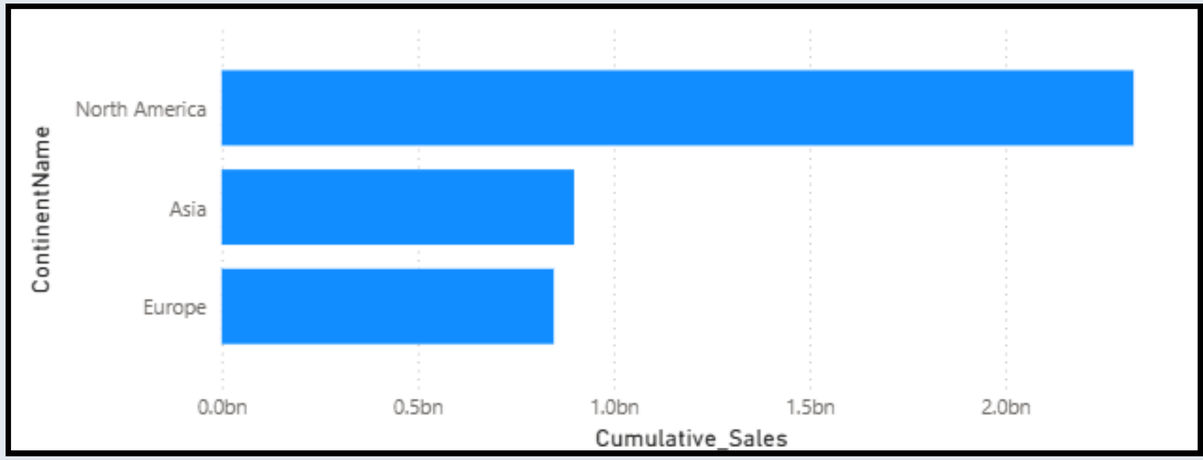
SALES TRENDS

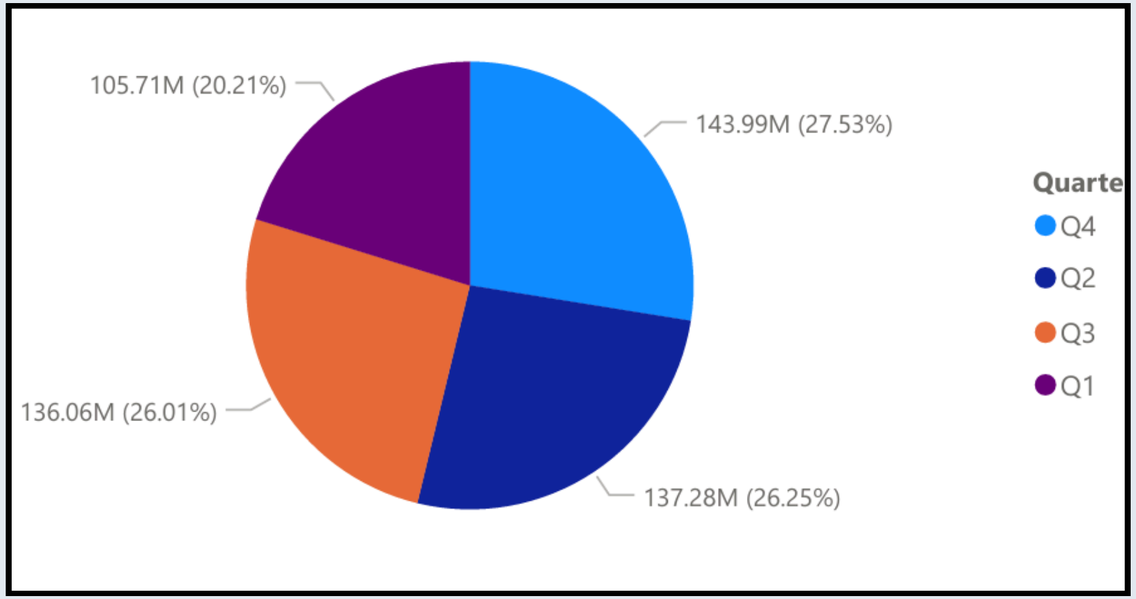
Sales Across Months

Month	Sum of TotalSales
November	50,206,705.89
May	47,273,928.48
December	47,027,747.56
October	46,760,409.70
July	46,759,349.44
August	45,784,254.89
June	45,374,835.60
April	44,632,060.76
September	43,516,271.59
March	36,274,936.44
February	35,602,044.33
January	33,836,141.17

Sales Across Regions



Sales Across Quarters



Sales Across Channels

ChannelDescription	Total_Sales
Store	2,395,539,542.80
Online	1,078,692,419.66
Reseller	789,185,269.84
Catalog	587,595,341.32

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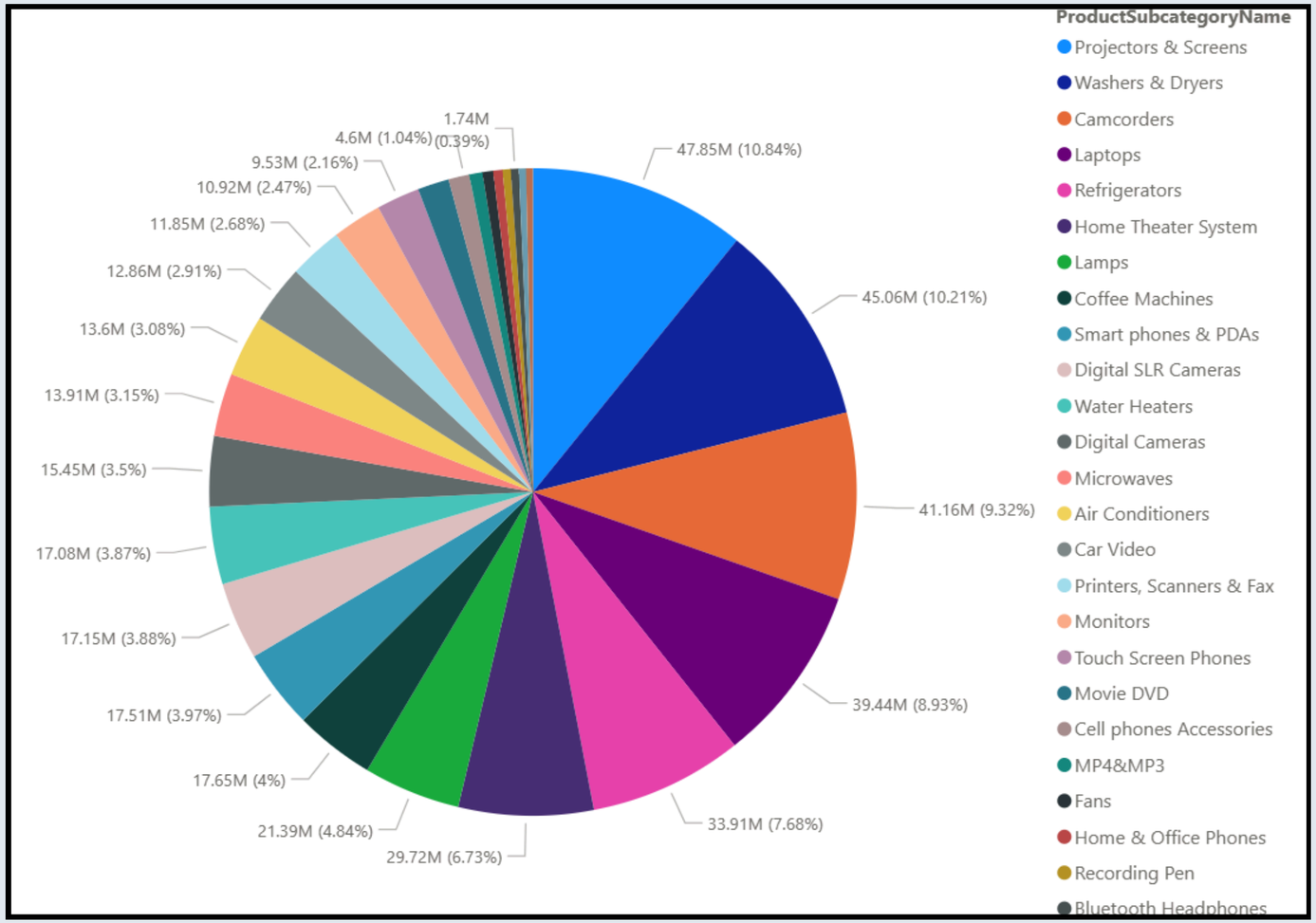
Highest Paying Customers

CustomerName	Sum of AOV
Trinity Richardson	75,151.25
Jasmine Jackson	70,413.00
Corey Raji	63,999.80
Natalie Jenkins	54,540.00
Evan Evans	54,276.75
Christian Perry	53,058.00
Jack Li	53,058.00
Julian Griffin	53,058.00

Product Sales



Product Sales:

- Revenue distribution is uneven, with certain product categories consistently outperforming others.
- High-value or trending products contribute disproportionately to total sales.

SHighest Paying Customer:

- Top 2 customers are females with the highest AOV. Some stores consistently exceed targets, while others underperform,

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```
use contoso_dataset;

-- products that generate the highest revenue
select p.ProductKey, p.ProductName, sum(f.SalesAmount) as Total_Revenue
from factsales f
join dimproduct p on f.ProductKey = p.ProductKey
group by p.ProductKey, p.ProductName
order by Total_Revenue desc;

-- Seasonal performance of top products
select p.productkey, p.productname, d.monthnumber as month, sum(f.salesamount) as monthly_revenue
from factsales f
join dimproduct p on f.productkey = p.productkey
join dimdate d on f.datekey = d.datekey
group by p.productkey, p.productname, month
order by monthly_revenue desc;


-- Regions generating highest sales for each product
select p.productname, g.regioncountryname, sum(f.salesamount) as total_revenue
from factsales f
join dimproduct p on f.productkey = p.productkey
join dimstore s on f.storekey = s.storekey
join dimgeography g on s.geographykey = g.geographykey
group by p.productname, g.regioncountryname
order by total_revenue desc;
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- top customers by revenue
select c.CustomerKey, c.FirstName, c.LastName, sum(f.SalesAmount) as Total_Revenue
from factonlinesales f
join dimcustomer c on f.CustomerKey = c.CustomerKey
group by c.CustomerKey, c.FirstName, c.LastName
order by Total_Revenue desc;
-- Total spending by product category per customer
select c.customerkey, c.firstname, c.lastname, pc.productcategoryname, sum(f.salesamount) as category_spend
from factonlinesales f
join dimcustomer c on f.customerkey = c.customerkey
join dimproduct p on f.productkey = p.productkey
join dimproductsubcategory ps on p.productsubcategorykey = ps.productsubcategorykey
join dimproductcategory pc on ps.productcategorykey = pc.productcategorykey
group by c.customerkey, c.firstname, c.lastname, pc.productcategoryname
order by c.customerkey, category_spend desc;
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-- Monthly spending trend per customer
select c.customerkey, c.firstname, c.lastname, d.monthnumber as month, sum(f.salesamount) as monthly_spend
from factonlinesales f
join dimcustomer c on f.customerkey = c.customerkey
join dimdate d on f.datekey = d.datekey
group by c.customerkey, c.firstname, c.lastname, month
order by c.customerkey, month;
```

ONLINE SALES TREND

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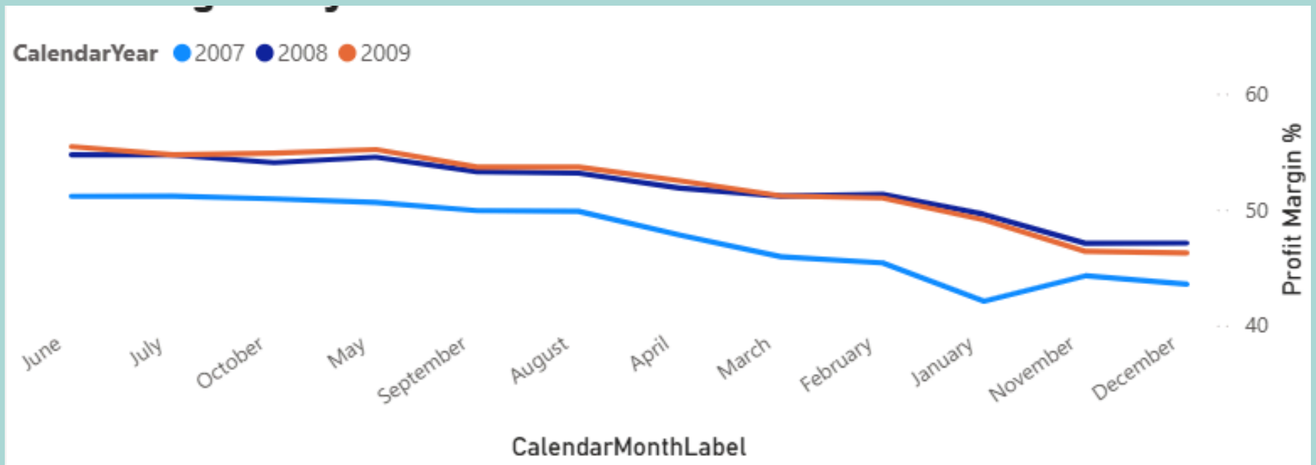
Profit Margin of Each Product Sold Online

ProductName	BrandName	Profit Margin %	Total Profit
Adventure Works 52" LCD HDTV X590 Brown"	Adventure Works	65.06	735,460.91
Contoso Bluetooth Active Headphones L15 Black	Contoso	64.58	1,085,874.77
A. Datum Full Frame Digital Camera X300 Green	A. Datum	64.30	99,236.43
Litware Home Theater System 7.1 Channel X711 Black	Litware	64.21	262,318.46
Contoso Telephoto Conversion Lens X400 Black	Contoso	64.19	251,283.96
Adventure Works LCD24 X300 White	Adventure Works	64.15	237,881.70
Fabrikam Refrigerator 24.7CuFt X9800 Grey	Fabrikam	64.11	1,617,187.72
Contoso Macro Zoom Lens X300 Silver	Contoso	64.07	148,565.46
Fabrikam Refrigerator 24.7CuFt X9800 White	Fabrikam	63.99	1,335,625.05
Contoso Bluetooth Active Headphones L15 Red	Contoso	63.97	1,107,951.83
Contoso Telephoto Conversion Lens X400 White	Contoso	63.96	219,036.66
Contoso Washer & Dryer 27in L270 White	Contoso	63.95	759,154.30
The Phone Company PDA Palm 4.7 inch L850 White	The Phone Company	63.92	241,070.88
Proseware LCD24W X300 Black	Proseware	63.91	328,375.77
Fabrikam Refrigerator 24.7CuFt X9800 Green	Fabrikam	63.91	1,460,741.17
Fabrikam Refrigerator 24.7CuFt X9800 Silver	Fabrikam	63.90	1,297,013.18
Contoso Telephoto Conversion Lens X400 Silver	Contoso	63.90	39,381,885.52
Total		**50.55**	**1,373,946,187.19**



Total Online Sales

2.72bn

Total Sales

Total Cost

1.34bn

Total Cost

Total Online Profit

1.37bn

Total Profit

Online Profit Margin

50.55

Profit Margin %

ANALYSIS GOALS:

1) We want to know which products make the most profit relative to their sales, not just which sell the most.

2) Instead of focusing on products, the goal here is to understand how profitability changes over time.

RESULTS:

- Total online sales reached 2.72bn with 1.37bn profit, resulting in a strong 50.55% profit margin.
- High-margin products dominate online sales, with most top items achieving 63–65% profit margins.
- Adventure Works leads in profit margin, while Fabrikam generates high total profit through strong sales volume.
- Contoso consistently contributes across multiple profitable product categories.
- Monthly profit margins are highest during mid-year (May–July) and decline towards year-end due to seasonal discounts.
- Overall profitability improved over time, indicating better pricing strategies and cost efficiency.

RECOMMENDATIONS

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Customer-Related

- Low customer retention rate
- Customer base is largely professional, aligning with relatively high AOV
- Even gender split → gender-neutral targeting is appropriate
- Some regions show high AOV despite lower sales volume → focus on premium products and branding
- Prioritize retention, upselling, and bundles to increase revenue per customer

Sales & Profitability

- Q4 drives highest sales, while mid-year (May–July) shows strongest profit margins
- Year-end margin decline linked to seasonal discounting
- Online channel performs strongly (2.72bn sales, 50.55% profit margin)
- High-margin products dominate online sales (≈63–65% margins)
- Adventure Works leads in profit margin; Fabrikam excels through high sales volume
- Contoso products show consistent profitability across multiple categories
- Overall profitability improves over time → evidence of better pricing and cost efficiency

Key Insights:

- Refocus product strategy on high-demand, fast-moving electronics and bundles, while reducing exposure to slow-selling large appliances.
- Strengthen digital and high-performing channels (especially online) and optimize or scale back underperforming sales channels.
- Use targeted regional pricing and promotions to stimulate demand and improve customer retention in key markets.

Thank you very much!